Supplementary Methodology

Machine-Learning-Guided Discovery of Oxide-Aperture Geometries for Polarization-Switching VCSEL Ising Nodes

Polina Morozova

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Abstract

This supplementary document describes the computational design loop used in the accompanying project notebook. The objective is to use machine learning as an active design engine to discover an oxide-aperture geometry that supports a clean, current-driven transition between two orthogonal VCSEL polarization states. A reduced-order polarization simulator provides the physics evaluation, while Gaussian Process Regression (GPR) with a Matérn 5/2 automatic-relevance-determination (ARD) kernel learns the geometry-to-switching-utility map. Expected Improvement (EI) uses the posterior mean and uncertainty to select the next aperture for simulation. The workflow therefore concentrates expensive physics evaluations on geometries that are both promising and informative. The resulting candidate can subsequently be tested with a higher-fidelity full-wave solver and experimentally fabricated.

1 Research objective and design philosophy

The target is not simply to maximize polarization splitting. A large built-in anisotropy can lock a VCSEL into one preferred polarization, whereas a polarization-encoded Ising node benefits from access to two competing states. Recent experiments on tailored oxide apertures show that reducing or compensating the intrinsic polarization preference can promote deterministic polarization switching and can reduce the optical injection power required for polarization locking in VCSEL Ising architectures [2, 3, 4].

The computational question is therefore:

> **Can uncertainty-aware machine learning discover an oxide-aperture geometry that produces a clean and robust current-driven polarization transition using substantially fewer physics evaluations than blind search?**

The primary output of the optimization is a **geometry**. Surrogate accuracy is evaluated as a secondary diagnostic because the surrogate must be sufficiently reliable to drive the search. Physical interpretation is then used to understand which geometric descriptors are associated with the discovered design.

2 Continuous aperture representation

Rather than restricting the search to a short list of named shapes, the aperture boundary is parameterized by a low-order Fourier series,

$$r(\phi) = R\left[1 + \sum_{m\in\{2,4,6\}} a_m \cos(m\phi - m\theta) + b_m \sin(m\phi - m\theta)\right]. \tag{1}$$

The implementation uses six Fourier coefficients and an in-plane rotation angle θ. The enclosed area is normalized after generating the boundary so that the learning problem is focused primarily on *shape* rather than trivial area scaling.

The bounded coefficients are chosen to avoid strongly self-intersecting or collapsed boundaries. Candidate geometries are filtered using minimum and maximum radial bounds and a roughness criterion. The search space consequently contains conventional forms such as circles and ellipses but also smooth, non-standard shapes that are not specified in advance.

For each aperture, a compact descriptor vector is constructed from:

$$\boldsymbol{x} = [A_2, A_4, A_6, Q, \theta, P_{\mathrm{norm}}, \rho_r, A_{\mathrm{norm}}, \sin 2\theta, \cos 2\theta], \tag{2}$$

where A_2, A_4, A_6 are Fourier amplitudes, Q is a second-order anisotropy measure obtained from the boundary second-moment tensor, θ is the inferred principal-axis orientation, P_{norm} is the normalized perimeter, and ρ_r measures radial roughness.

3 Reduced-order VCSEL polarization simulator

The prototype physics backend follows the established spin-flip-model (SFM) description of VCSEL polarization dynamics. The SFM accounts for the two orthogonal complex electric-field components together with total carrier inversion and spin-population imbalance, and includes linear dichroism, linear birefringence, linewidth enhancement, and spin-flip relaxation [1].

The simulator propagates

$$E_x, \quad E_y, \quad D, \quad n \tag{3}$$

over normalized bias current

$$\mu = \frac{I}{I_{\mathrm{th}}}. \tag{4}$$

For the present proof-of-concept, the normalized current is sampled over $1.05 \le \mu \le 2.05$.

Using the convention implemented in the notebook, the complex field equations are

$$\dot{E}_x = -(\kappa + \gamma_a)E_x - i(\kappa\alpha + \gamma_p)E_x + \kappa(1 + i\alpha)(DE_x + inE_y), \tag{5}$$

$$\dot{E}_y = -(\kappa - \gamma_a)E_y - i(\kappa\alpha - \gamma_p)E_y + \kappa(1 + i\alpha)(DE_y - inE_x), \tag{6}$$

with carrier equations

$$\dot{D} = -\gamma\left[D(1 + |E_x|^2 + |E_y|^2) - \mu + \Re(inC)\right], \tag{7}$$

$$\dot{n} = -\gamma_s n - \gamma\left[n(|E_x|^2 + |E_y|^2) + \Re(iDC)\right], \tag{8}$$

where

$$C = E_y E_x^* - E_x E_y^*. \tag{9}$$

The parameters γ_a and γ_p represent linear amplitude anisotropy (dichroism) and linear phase anisotropy (birefringence), respectively. The aperture geometry contributes a structural-anisotropy term derived from the second-moment tensor. In the prototype, this contribution is represented phenomenologically as

$$\gamma_a^{(0)} = \gamma_{a,\mathrm{nat}} + c_a Q\cos(2\theta), \tag{10}$$

$$\gamma_p = \gamma_{p,0} + c_p\left[\frac{Q}{2} + 0.15|\sin(2\theta)|\right]. \tag{11}$$

The numerical coefficients are normalized proof-of-concept parameters, not experimentally fitted constants for a particular wafer. They are kept explicit so that this geometry-to-anisotropy map can later be replaced by measured or full-wave data without changing the ML layer.

4 Switching utility used by the optimizer

At each current value the polarization metric is calculated as

$$P(\mu) = \frac{I_x - I_y}{I_x + I_y}, \tag{12}$$

where $I_x = |E_x|^2$ and $I_y = |E_y|^2$. A useful switching design should show a substantial decrease in P over a relatively narrow current interval, with the polarization changing from predominantly x- to predominantly y-dominated emission.

The continuous switching utility used by the prototype combines four considerations:

1. magnitude of the polarization swing,
2. sharpness of the transition,
3. proximity of the transition to a target normalized-current window,
4. a penalty for excessive geometric complexity.

This continuous objective is preferable to a binary "switch/no-switch" label because it gives Bayesian optimization a smoother quantity to model. The present score is a design utility rather than a universal physical metric; for an experimental study it should be replaced or recalibrated using the experimentally relevant switching criterion.

5 Machine-learning component: GPR + Expected Improvement

The central ML mechanism is a Gaussian-process surrogate,

$$f(\boldsymbol{x}) \sim \mathcal{GP}\left(m(\boldsymbol{x}), k(\boldsymbol{x}, \boldsymbol{x}')\right), \tag{13}$$

with a Matérn 5/2 ARD kernel. ARD assigns an independent characteristic length scale to each geometric descriptor, allowing the model to learn which directions in design space produce the fastest changes in switching utility.

The Matérn kernel used is

$$k(r) = \sigma_f^2 \left(1 + \frac{\sqrt{5}r}{\ell} + \frac{5r^2}{3\ell^2}\right) \exp\left(-\frac{\sqrt{5}r}{\ell}\right), \tag{14}$$

with the ARD distance defined by the feature-specific length scales ℓ_j.

The important role of the GPR is not simply prediction. For an unevaluated candidate the model returns both a posterior mean $\mu(\boldsymbol{x})$ and predictive standard deviation $\sigma(\boldsymbol{x})$. These are inserted into the Expected Improvement acquisition function,

$$EI(\boldsymbol{x}) = (\mu(\boldsymbol{x}) - f^* - \xi)\Phi(z) + \sigma(\boldsymbol{x})\phi(z), \tag{15}$$

where

$$z = \frac{\mu(\boldsymbol{x}) - f^* - \xi}{\sigma(\boldsymbol{x})}, \tag{16}$$

f^* is the best observed utility, ξ is a small exploration parameter, and Φ and ϕ are the standard normal cumulative distribution and density functions.

The next geometry is

$$\boldsymbol{x}_{\text{next}} = \arg\max_{\boldsymbol{x} \in \mathcal{P} \setminus \mathcal{O}} EI(\boldsymbol{x}), \tag{17}$$

where $\mathcal{P}$ is the candidate pool and $\mathcal{O}$ is the set of already evaluated geometries.

This is the key design principle of the project: **ML determines where the next expensive physics calculation should be spent**. The algorithm can therefore exploit regions where the predicted switching utility is high while exploring regions where the uncertainty is high.

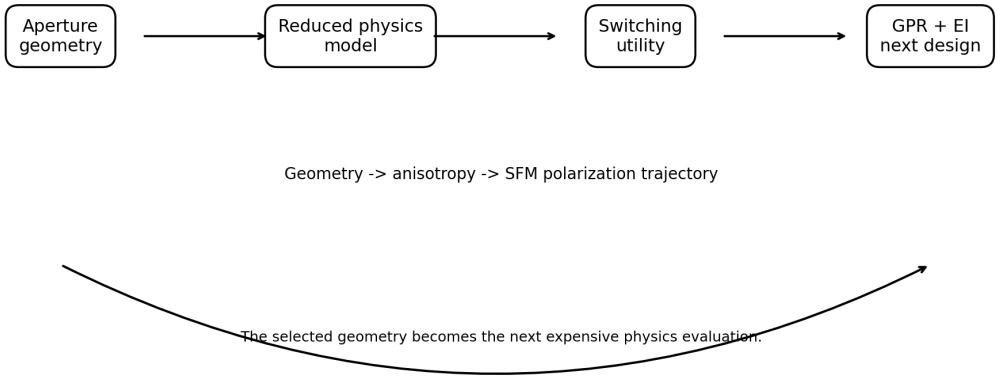


Figure 1: Conceptual workflow: a continuous aperture geometry is evaluated by the physics backend, reduced to a switching utility, learned by a GPR surrogate, and used through Expected Improvement to select the next geometry. The final candidate is intended for subsequent high-fidelity simulation and experimental validation.

6 Sequential Bayesian-optimization workflow

The candidate pool is generated once using Latin-hypercube sampling of the bounded Fourier parameters. The same pool is used for Bayesian optimization and for the random-search baseline, so the comparison concerns search strategy rather than different candidate spaces.

The sequential workflow is:

1. Randomly select an initial set of apertures and evaluate them with the physics simulator.
2. Fit the GPR surrogate to the evaluated geometries and their switching utilities.
3. Predict the posterior mean and uncertainty for all untested geometries.
4. Compute Expected Improvement for every untested geometry.
5. Simulate the geometry with maximum EI and add its result to the training set.
6. Repeat until the evaluation budget is exhausted.

7 Comparison with blind search

A random-search baseline uses exactly the same candidate pool and evaluation budget but chooses every new geometry uniformly at random. The primary optimization diagnostic is the best-so-far switching utility as a function of the number of physics evaluations.

A stronger campaign-level evaluation should repeat the procedure over multiple random seeds and report the median together with a variability band. This tests whether the apparent advantage of Bayesian optimization is reproducible rather than a consequence of one favorable initialization.

The intended question is not whether the surrogate eventually reproduces every simulated point. It is whether the uncertainty-aware surrogate allows the optimizer to discover high-utility geometries with fewer expensive evaluations.

8 Surrogate validation

The surrogate is evaluated separately from the optimization result. Predictions should be made for data that were not used to fit the corresponding model. The notebook includes five-fold cross-validation on the evaluated BO observations and reports

$$\mathrm{MAE} = \frac{1}{N}\sum_i |y_i - \hat{y}_i|, \tag{18}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_i (y_i - \hat{y}_i)^2}, \tag{19}$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \tag{20}$$

Because the final scientific objective is discovery of a geometry, surrogate accuracy is treated as a *trust diagnostic* rather than the headline performance measure. In a higher-fidelity study, a reserved set of geometries should be excluded entirely from the BO campaign and used only as a final external test set.

The uncertainty predictions should also be checked for calibration: cases with larger predicted uncertainty should, on average, have larger absolute errors. This is particularly important near the boundary of the sampled design space, where extrapolation can be unreliable.

9 Physical interpretation

The model is interpreted through descriptors that correspond to concrete geometric quantities rather than through raw pixels. The most physically relevant quantities are expected to include second-order anisotropy magnitude, aperture orientation, and higher-order Fourier content. Permutation importance provides a simple diagnostic of how strongly each descriptor contributes to predictive performance.

Interpretation is deliberately separated from causality. A descriptor that is important to the surrogate identifies a useful statistical direction in the learned design landscape; it does not by itself prove that the corresponding geometric mechanism causes polarization switching. Physical conclusions should therefore be checked by controlled geometry perturbations, higher-fidelity simulations, or experiment.

10 Robustness and fabrication relevance

A selected aperture should not depend critically on an unrealistically precise boundary. For the candidate geometry, the prototype perturbs the Fourier coefficients and orientation and re-evaluates the switching utility. A robust design is one for which the useful switching behavior persists under small perturbations.

The same framework can be extended to fabrication-aware constraints such as minimum feature size, rounding limits, maximum perimeter, oxidation-radius tolerance, and allowed alignment relative to the wafer/crystal axes.

11 Connection to polarization-encoded photonic Ising computing

Photonic Ising machines can use orthogonal VCSEL polarization states as binary information states. In the VCSEL architecture studied by Yuan and co-workers, polarization locking under optical injection provides a mechanism for mapping interactions between nodes, while the

intrinsic polarization preference of a VCSEL can make this process more difficult [3, 2]. Their experiments show that tailored oxidation apertures can compensate intrinsic anisotropy and produce controllable polarization switching; the custom-aperture work further demonstrates current-dependent switching with non-standard aperture geometries [4].

The present project addresses the next design step: instead of manually selecting a small number of aperture shapes, the design space is searched continuously and adaptively. The ML system therefore operates at the device-design level, producing a candidate intended to make the polarization degree of freedom easier to control in a later array-level Ising implementation.

A future system-level experiment would insert the selected VCSEL parameters into a mutually injection-locked array model and evaluate computational metrics on QUBO or Max-Cut problems. That stage is outside the current prototype and is kept separate from the device-level aperture optimization.

12 Validation ladder for the physics backend

The physics simulator and the ML surrogate represent different validation problems. The reduced-order backend should progressively be replaced or calibrated through the following ladder:

1. Numerical sanity checks and convergence of the ODE integration.
2. Comparison with canonical SFM polarization behavior and published polarization-switching results [1].
3. Calibration of aperture-to-anisotropy coefficients using geometry-specific full-wave modal calculations or experiments.
4. Comparison against measured polarization-resolved spectra, near-field modes, switching currents, or related observables for the target device.
5. Only after these checks, use a larger physics-generated dataset as the basis for the ML discovery campaign.

This distinction matters because high surrogate accuracy can only establish agreement with the simulator. It cannot, by itself, establish agreement with the physical device.

13 Replacing the prototype backend with full-wave data

The ML interface is intentionally modular. A higher-fidelity workflow can replace the reduced-order geometry-to-anisotropy map with geometry-specific modal quantities exported from Lumerical, COMSOL, an FEM/eigenmode solver, or another validated electromagnetic tool. The ML stage can then operate on a table containing geometry descriptors and physical quantities such as polarization-resolved modal frequencies, modal losses, anisotropy parameters, and switching metrics.

The GPR and EI machinery does not need to change when the physics backend is upgraded. This separation allows the methodological question – whether uncertainty-aware active learning can efficiently search a large aperture design space – to remain intact while the physical model becomes progressively more realistic.

14 Current prototype boundary

The accompanying notebook is a computational proof-of-concept and methods prototype. Its reduced-order geometry-to-anisotropy coefficients are not extracted from the specific 940-nm

VCSEL wafer, and the selected candidate should not be described as a globally optimal fabricated device. The scientifically meaningful deliverable at this stage is the modular workflow:

continuous aperture family → polarization physics → switching utility → GPR uncertainty model → EI-selected geometry.

When full-wave or experimental calibration data become available, the physical backend can be substituted while preserving the ML design engine.

15 Reproducibility and code availability

The computational implementation, notebook, figure used in this document, and project metadata are provided in the accompanying repository:

`https://github.com/YOUR_GITHUB_USERNAME/vcsel-polarization-bo`

Replace `YOUR_GITHUB_USERNAME` with the actual GitHub account name before compiling the final supplementary PDF. The repository is intended to contain the notebook under `notebooks/`, this LaTeX source under `supplementary/`, and the workflow figure under `figures/`.

References

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